

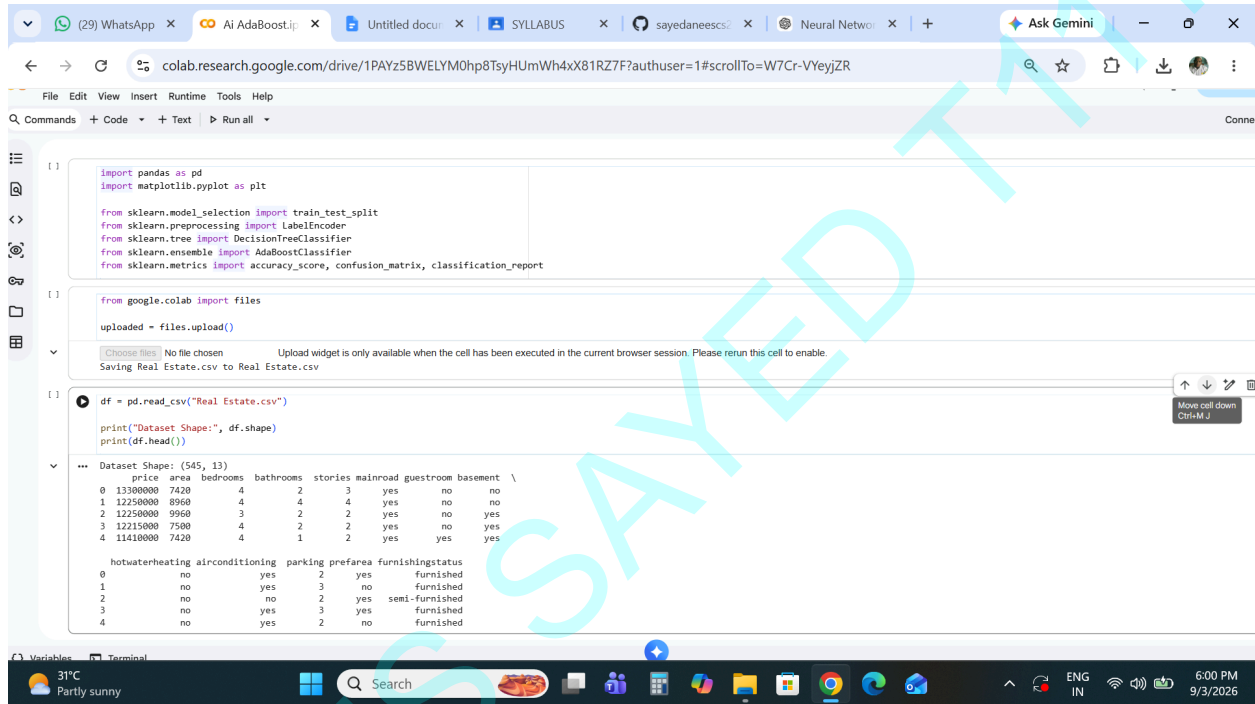
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SUBJECT:- AI

Practical 7

AIM:- Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.



```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
from google.colab import files
uploaded = files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Real Estate.csv to Real Estate.csv

```
df = pd.read_csv("Real Estate.csv")
print("Dataset Shape:", df.shape)
print(df.head())
```

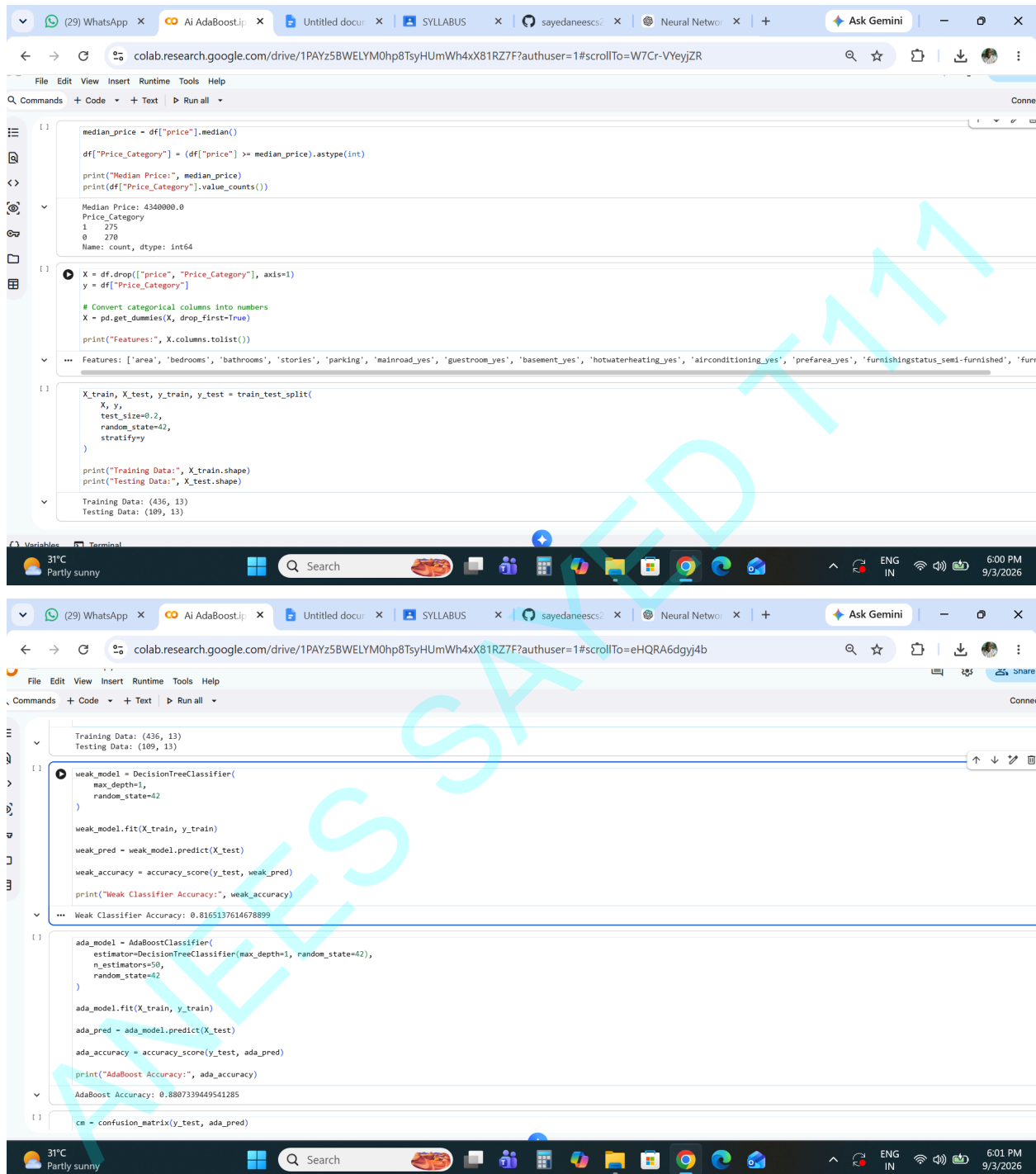
Dataset Shape: (545, 13)

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	\
0	13300000	7420	4	2	3	yes	no	no	
1	12250000	8960	4	4	4	yes	no	no	
2	12250000	9960	3	2	2	yes	no	yes	
3	12215000	7500	4	2	2	yes	no	yes	
4	11410000	7420	4	1	2	yes	yes	yes	

	hotwaterheating	airconditioning	parking	prefarea	furnishingsstatus
0	no	yes	2	yes	furnished
1	no	yes	3	no	furnished
2	no	no	2	yes	semi-furnished
3	no	yes	3	yes	furnished
4	no	yes	2	no	furnished

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The image displays two screenshots of a Google Colab notebook, showing Python code for data preprocessing and model training. A large, diagonal watermark reading "SAVED" is overlaid across both screenshots.

Top Screenshot: The notebook interface shows the following code cells:

```
median_price = df["price"].median()

df["Price_Category"] = (df["price"] >= median_price).astype(int)

print("Median Price:", median_price)
print(df["Price_Category"].value_counts())

# Output:
# Median Price: 4340000.0
# Price_Category
# 1    275
# 0     270
# Name: count, dtype: int64

X = df.drop(["price", "Price_Category"], axis=1)
y = df["Price_Category"]

# Convert categorical columns into numbers
X = pd.get_dummies(X, drop_first=True)

print("Features:", X.columns.tolist())

# Output:
# Features: ['area', 'bedrooms', 'bathrooms', 'stories', 'parking', 'mainroad_yes', 'guestroom_yes', 'basement_yes', 'hotwaterheating_yes', 'airconditioning_yes', 'prefarea_yes', 'furnishingstatus_semi-furnished', 'furnishingstatus_furnished']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)

# Output:
# Training Data: (436, 13)
# Testing Data: (109, 13)
```

Bottom Screenshot: The notebook continues with the following code cells:

```
weak_model = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

weak_model.fit(X_train, y_train)

weak_pred = weak_model.predict(X_test)

weak_accuracy = accuracy_score(y_test, weak_pred)

print("Weak Classifier Accuracy:", weak_accuracy)

# Output:
# Weak Classifier Accuracy: 0.8165137614678899

ada_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1, random_state=42),
    n_estimators=50,
    random_state=42
)

ada_model.fit(X_train, y_train)

ada_pred = ada_model.predict(X_test)

ada_accuracy = accuracy_score(y_test, ada_pred)

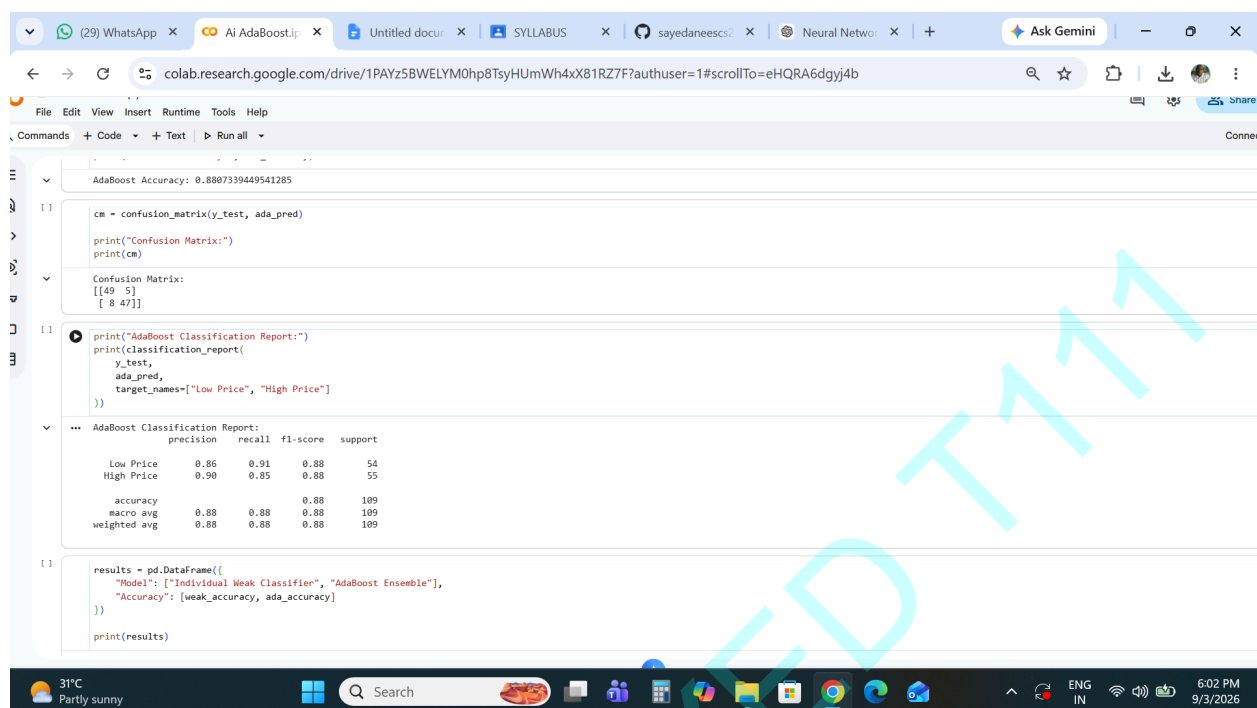
print("AdaBoost Accuracy:", ada_accuracy)

# Output:
# AdaBoost Accuracy: 0.8807339449541285

cm = confusion_matrix(y_test, ada_pred)
```

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The first screenshot shows a Google Colab notebook with the following code and output:

```
AdaBoost Accuracy: 0.8807339449541285
```

```
cm = confusion_matrix(y_test, ada_pred)
print("Confusion Matrix:")
print(cm)
```

Confusion Matrix:

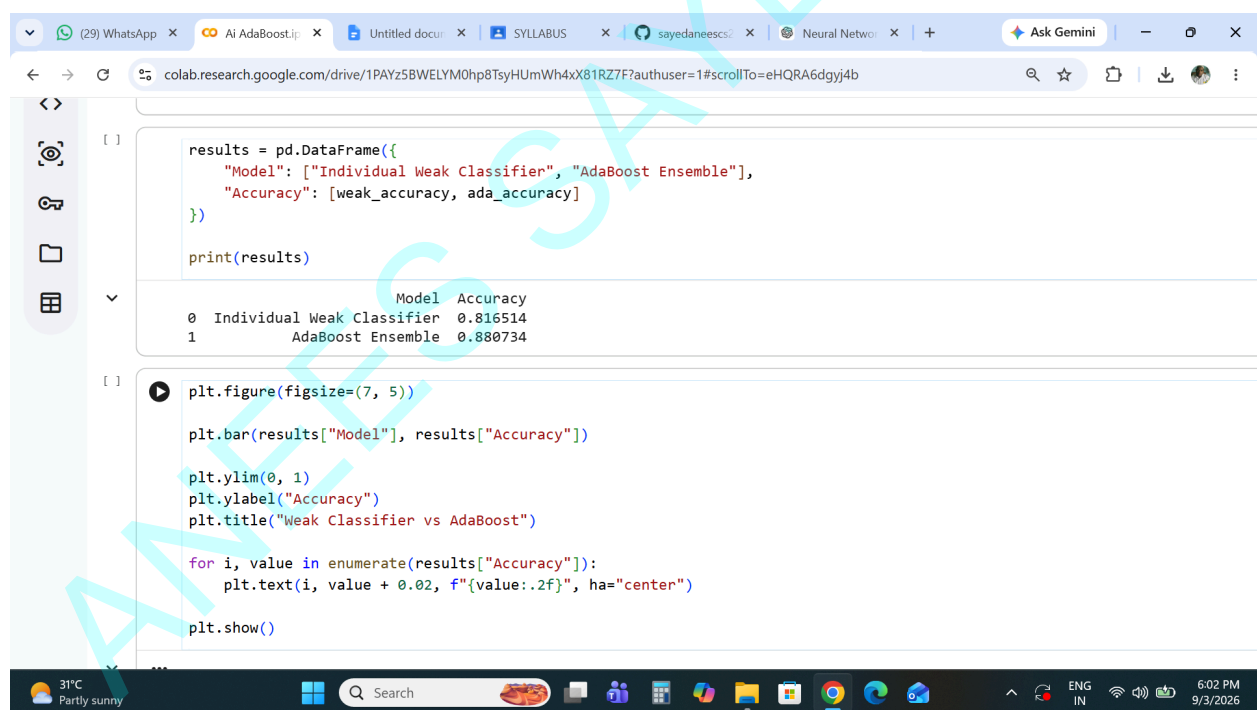
```
[[49  5]
 [ 8 47]]
```

```
print("AdaBoost Classification Report:")
print(classification_report(
    y_test,
    ada_pred,
    target_names=["Low Price", "High Price"]
))
```

AdaBoost Classification Report:

	precision	recall	f1-score	support
Low Price	0.86	0.91	0.88	54
High Price	0.90	0.85	0.88	55
accuracy			0.88	109
macro avg	0.88	0.88	0.88	109
weighted avg	0.88	0.88	0.88	109

```
results = pd.DataFrame({
    "Model": ["Individual Weak Classifier", "AdaBoost Ensemble"],
    "Accuracy": [weak_accuracy, ada_accuracy]
})
print(results)
```



The second screenshot shows the same notebook with the following code and output:

```
results = pd.DataFrame({
    "Model": ["Individual Weak Classifier", "AdaBoost Ensemble"],
    "Accuracy": [weak_accuracy, ada_accuracy]
})
print(results)
```

Model Accuracy

	Model	Accuracy
0	Individual Weak Classifier	0.816514
1	AdaBoost Ensemble	0.880734

```
plt.figure(figsize=(7, 5))
plt.bar(results["Model"], results["Accuracy"])
plt.ylim(0, 1)
plt.ylabel("Accuracy")
plt.title("Weak Classifier vs AdaBoost")
for i, value in enumerate(results["Accuracy"]):
    plt.text(i, value + 0.02, f"{value:.2f}", ha="center")
plt.show()
```

The bar chart displays the accuracy of two models: Individual Weak Classifier (0.816514) and AdaBoost Ensemble (0.880734). The y-axis is labeled 'Accuracy' and ranges from 0 to 1. The x-axis is labeled 'Model'.

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